10TH CLASS CHEMISTRY — NEW SYLLABUS 2025-26 (PECTAA)

Chapter 26 — Polymers

Solved Quick Checks + Exercise | New Book (PECTAA 2025-26)

Quick Checks (Solved)

26.1 Give an example of a commercial product which is formed using polyethene.

Polyethene (polythene) is used to make shopping bags, carry-home food containers, soft drink bottles, water bottles, juice bottles, kitchen-wares, buckets, food wrappers, cups and plates.

26.1 Write three properties of synthetic polymers.

(i) They are light weight, flexible and durable.

(ii) They are strong and resistant to heat and chemicals, and they do not shrink or wrinkle.

(iii) They can be easily moulded into solid objects of almost any desired shape, which is why they are used in almost every sphere of our daily life.

26.2 How does condensation polymerization occur?

A condensation polymer is formed when two different molecules containing two different functional groups react together with the elimination of a small molecule like H_2O, HCl or NH_3. The reaction takes place in the presence of an acid or a base. Since the functional groups are present at both ends of the resulting molecule, the polymerization continues at both ends and a very big molecule is formed.

26.2 How an ester linkage between two monomers is formed?

An ester linkage is formed when the carboxyl group (–COOH) of a dicarboxylic acid reacts with the hydroxyl group (–OH) of a diol. A molecule of water is eliminated and the linkage –CO–O– is formed between the two monomers. For example, ethane diol reacts with terephthalic acid in this way and the polymer formed is PET (polyethylene terephthalate).

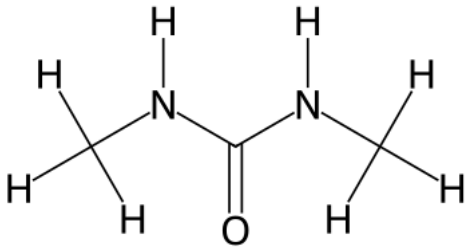
26.3 What do you understand by the term repeating unit?

The repeating unit is the smallest structural part which is repeated again and again in a polymer chain. It is written inside brackets with the subscript n, where n shows the large number of times the unit is repeated. For example, the repeating unit of polyethene is $–CH_2–CH_2–$ and that of PET is $–CO–C_6H_4–CO–O–CH_2–CH_2–O–$.

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26.3 Draw the structure of a polyamide linkage.

In a polyamide, the dicarboxylic acid and the diamine are joined by amide linkages. The polyamide linkage is –CO–NH–, which is formed with the loss of a molecule of water from a –COOH group and an $–NH_2$ group.



Exercise (Solved)

A. Multiple Choice Questions

1. Which of the following polymer is a synthetic polymer:

(a) Starch (b) Cellulose

(c) Animal fat (d) Polyester

Answer: (d) — Polyester is a man-made polymer prepared from a diol and a dicarboxylic acid. Starch and cellulose are natural polymers of plants and animal fat is a natural compound.

2. Which polymer has an amide linkage:

(a) Polyester (b) Polyamide

(c) Polyethene (d) Polystyrene

Answer: (b) — Polyamide is formed when a dicarboxylic acid reacts with a diamine, and the linkage formed between the monomers is the amide linkage –CO–NH–. Polyester contains an ester linkage and polyethene has only C–C bonds.

3. Identify an ester linkage:

(a) R–CO–O–R (b) R–CO–R

(c) R–CO–O–NH–R (d) R–CO–NH–R

Answer: (a) — The ester linkage is –CO–O–, which is formed when a carboxyl group reacts with a hydroxyl group. Option (d) is the amide linkage of a polyamide.

4. The structure of the monomer of the following polymer is $[-CH_2-CH(OCOCH_3)-]_n$

(a) $CH_2=CH_2$ (b) $CH_2=CH–CH_3$

(c) $CH_2=CH–OCOCH_3$ (d) $CH_2=CH–CO–O–CH_3$

Answer: (c) — To get the monomer from an addition polymer, the brackets, the extended bonds and the subscript n are removed and the single bond of the chain is changed into a double bond. This gives vinyl acetate, $CH_2=CH–OCOCH_3$.

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5. Which of the following polymers can be recycled?

(a) PET (b) Polystyrene

(c) PVC (d) Epoxy

Answer: (a) — PET can be recycled by chemically breaking it down into its monomer units. Chemical recycling can be done by heating the polymer in the absence of oxygen, and the monomers obtained can be polymerised again.

6. Which of the following polymers is biodegradable?

(a) Polyamide (b) Polyester

(c) Starch (d) PET

Answer: (c) — Starch is a natural polymer, so it undergoes biodegradation by the action of microorganisms. Unlike this, plastics such as polyamide, polyester and PET do not undergo biodegradation and can take up to 1000 years to break down.

7. Which of the following polymers is obtained by condensation polymerization?

(a) Polyvinyl chloride (b) Polystyrene

(c) Polyamide (d) Polyethene

Answer: (c) — Polyamide is formed by the condensation of a dicarboxylic acid with a diamine with the elimination of water, so it is a condensation polymer. The other three are addition polymers.

B. Short Answer Questions

26.1 Give two examples each of addition polymers and condensation polymers.

Addition polymers: polyethene (polythene) and polystyrene; polyvinyl chloride (PVC) and polypropene are also addition polymers.

Condensation polymers: polyester (PET) and polyamide (nylon); bakelite is also a condensation polymer.

26.2 How polymers are useful for us?

Polymers are very useful because they have high strength, flexibility and resistance to heat and chemicals, and they are light in weight. They are used to make carry-home food containers, soft drink and water bottles, shopping bags, kitchen-wares, buckets, tables, chairs, food wrappers, cups and plates. Synthetic fibres such as nylon and polyester are used in the textile industry, and polymers are also used in pipes, paints, adhesives and medical items.

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26.3 How synthetic fibers are different from natural fibers?

Synthetic fibres	Natural fibres
They are made by chemical polymerization in industry.	They are obtained from plants or animals, for example cotton, silk and wool.
They are cheaper in cost.	They are comparatively expensive.
They are stronger, more durable, light weight and quick drying.	They are comparatively weaker and take more time to dry.
They do not shrink and they resist wrinkling.	They shrink and wrinkle easily.
They are moth and insect resistant.	They are attacked by moths and insects.

26.4 Are natural polymers hazardous for the environment? Explain why?

Natural polymers are not as hazardous as synthetic polymers, because they undergo biodegradation. They are broken down by the action of microorganisms and bacteria, so they do not accumulate in the environment for a long time. For example, starch and cellulose rot away in a short time. However, if they are produced in very large amounts and are not disposed of properly, they can still cause some pollution.

26.5 Draw the structures of the following polymers: (a) Polypropylene (b) Polystyrene

(a) Polypropylene: it is an addition polymer of propene, so its repeating unit has a methyl group on every second carbon atom of the chain.

$$[\ -CH_2-CH(CH_3)-\ ]_n$$

(b) Polystyrene: it is an addition polymer of styrene, so a phenyl group (C_6H_5) is attached to every second carbon atom of the chain.

$$[\ -CH_2-CH(C_6H_5)-\ ]_n$$

In both cases n is a very large number, which shows that many monomer molecules have joined to form one big molecule.

C. Constructed Response Questions

26.1 Why is it dangerous to burn discarded articles made of plastics?

Burning of discarded plastic articles in towns creates a hazardous environmental condition, because this releases dangerous gases in the air. Plastics contain carbon and other elements, so their burning produces carbon monoxide, carbon dioxide, smoke and other toxic gases which pollute the air and cause serious health problems such as breathing difficulty and eye irritation. PVC also contains chlorine, so its burning can produce hydrogen chloride gas, which is corrosive. Therefore plastic waste should be recycled instead of being burnt.

26.2 Why is it not advisable to dump discarded synthetic articles in oceans?

Synthetic polymers do not undergo biodegradation, so articles made of plastics can take up to 1000 years to break down. If they are dumped in oceans, they remain there for a very long time and pollute the water. Aquatic animals such as fish and turtles swallow pieces of plastic or become trapped in them, and many of them die. The plastic slowly breaks into very small particles which mix with the water and enter the food chain. Therefore discarded synthetic articles should never be dumped in oceans, rivers or drains; they should be recycled or disposed of properly.

26.3 What is the role of catalyst in addition polymerization?

In addition polymerization, the double bond of the monomer must break so that the monomers can join together. The catalyst helps this bond to break and speeds up the joining of the monomer molecules, so the polymerization takes place at a suitable temperature and pressure. Polymerization reactions usually take place at high pressure and in the presence of a catalyst. Without the catalyst the reaction would be very slow.

26.4 Deduce the monomer from the following addition polymers: (i) $[-CH_2-CH(Cl)-]_n$ (ii) $[-CH_2-CH(CN)-]_n$

The brackets, the extended bonds at both ends and the subscript n are removed, and the single bond between the two carbon atoms of the repeating unit is changed into a double bond. In this way the monomer is obtained.

(i) The monomer is vinyl chloride (chloroethene), $CH_2=CHCl$, and the polymer is PVC (polyvinyl chloride).

(ii) The monomer is acrylonitrile, $CH_2=CHCN$, and the polymer is polyacrylonitrile.

$$n\ CH_2=CHCl \rightarrow [\ -CH_2-CH(Cl)-\ ]_n$$

$$n\ CH_2=CHCN \rightarrow [\ -CH_2-CH(CN)-\ ]_n$$

D. Descriptive Questions

26.1 How is PET formed? Can you recycle this polymer?

PET (polyethylene terephthalate) is a condensation polymer. It is formed when ethane diol (a diol) reacts with terephthalic acid (a dicarboxylic acid). The carboxyl group of terephthalic acid reacts with the hydroxyl group of the diol, a molecule of water is eliminated, and an ester linkage is formed. The product is called a repeat unit, and since functional groups are present at both of its ends, the polymerization continues at both ends to give a very big molecule.

$$n\ HO-CH_2-CH_2-OH + n\ HOOC-C_6H_4-COOH \rightarrow$$

$$[\ -O-CH_2-CH_2-O-CO-C_6H_4-CO-\ ]_n + 2n\ H_2O$$

Yes, PET can be recycled. It is collected, cleaned and chemically broken down into its monomer units; chemical recycling can be done by heating the polymer in the absence of oxygen. The monomers obtained are then purified and polymerised again to make new PET products.

26.2 Differentiate between addition and condensation polymerizations.

Addition polymerization	Condensation polymerization
A single monomer is used which has a double or a triple bond.	Two different monomers are used, and each monomer has two functional groups, one on each end.
It involves an addition reaction and only a single product is formed.	It involves condensation with the elimination of a small molecule like H_2O, HCl or NH_3.
No molecule is eliminated and no by-product is formed.	A small molecule is eliminated at every step.
The polymer formed contains only single bonds along the backbone chain.	The polymer formed contains a specific linkage, for example an ester or an amide linkage.
Its molecular mass can be found by adding the molecular masses of the monomers.	Its molecular mass cannot be found simply by adding the masses of the monomers.
Examples: polyethene, polystyrene, PVC.	Examples: polyester, nylon (polyamide), bakelite.

26.3 Write down reactions alongwith the conditions to prepare polyamide and polyester.

Polyester: it is prepared by the condensation of a diol with a dicarboxylic acid. For example, ethane diol reacts with terephthalic acid and an ester linkage is formed with the elimination of water. The reaction is carried out by heating the monomers, often with a catalyst, and the water formed is removed continuously.

$$n\ HO–CH_2–CH_2–OH + n\ HOOC–C_6H_4–COOH \rightarrow$$
$$[\ –O–CH_2–CH_2–O–CO–C_6H_4–CO–\ ]_n + 2n\ H_2O$$

Polyamide: it is formed when a dicarboxylic acid reacts with a diamine. Each –COOH group of the acid reacts with each $–NH_2$ group of the diamine, an amide linkage is formed and a water molecule is lost. For example, butanedioic acid reacts with butane-1,4-diamine on heating.

$$n\ HOOC–CH_2–CH_2–COOH + n\ H_2N–(CH_2)_4–NH_2 \rightarrow$$
$$[\ –CO–CH_2–CH_2–CO–NH–(CH_2)_4–NH–\ ]_n + 2n\ H_2O$$

Both reactions are carried out in the presence of an acid or a base, and the small molecule formed is removed so that the polymerization continues.

26.4 Describe the role of synthetic polymers in the textile industry.

Polymers which are used in the textile industry are called synthetic fibres; examples are nylon, polyester, rayon and acrylic. Their properties make them very useful in the textile industry:

1. They are cheaper in cost than natural fibres like cotton and silk.
2. They are stronger and more durable than natural fibres.
3. They do not shrink and they resist wrinkling.
4. They are light weight and quick drying, and they are moth and insect resistant.

Polyester and nylon are the two common polymers which account for 69 % of all the material used in the clothing industry worldwide. Polyester is an important fibre for its strength and resistance to shrinking, and it is commonly used in sportswear and hosiery. Acrylic is light weight, soft and provides warmth, so it is used in winter clothing. In this way synthetic polymers have replaced a large part of cotton and wool.

26.5 Describe the effects of synthetic polymers on the environment.

(i) Non-biodegradable waste: unlike other materials, plastics do not undergo biodegradation, and articles made of plastics can take up to 1000 years to break down. This has led to the build-up of huge amounts of plastic waste on land and in water.

(ii) Dumping in landfill sites and drains: discarded plastic articles are burnt, dumped in landfill sites or thrown in drains, from where they make their way to rivers and oceans. Plastic pollution is now found everywhere — on lands, in air and in all water bodies — and it has reached an alarming level.

(iii) Harm to aquatic life: in oceans, fish and other animals swallow pieces of plastic or become trapped in them. These plastics slowly break into very small particles which enter the food chain and reach human beings.

(iv) Toxic gases on burning: burning of plastic waste releases carbon monoxide, smoke and other dangerous gases into the air, which pollute the air and cause serious health problems.

To reduce these effects, unnecessary plastic should be avoided, articles should be reused, recyclable polymers such as PET should be recycled, and the waste should be disposed of safely.

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